6.4210/6.4212
Robotic Manipulation

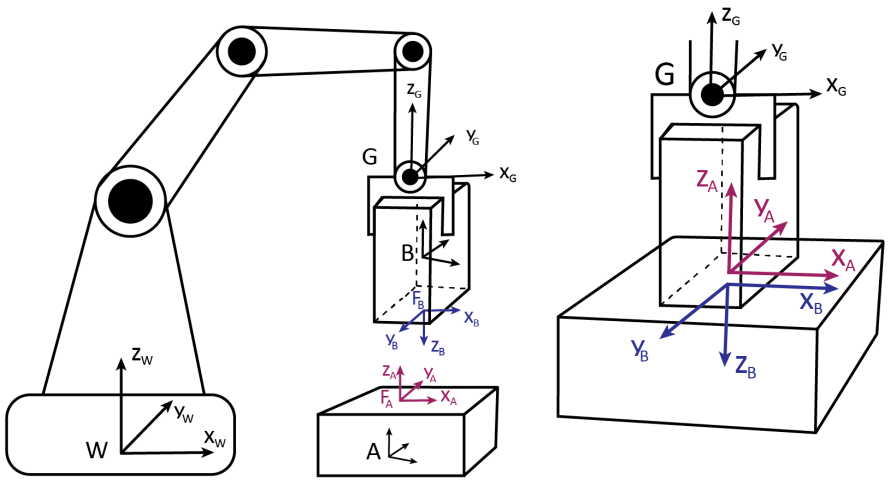
Midterm Practice

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6.4210/2
Exam Practice

1 Frames

1. In the figure on the left, the robot is holding block B in its gripper. We want to place the bottom face of B on the top face of block A so that the origin of the face frames coincide and the x axes are aligned (as shown on the right). For each object, we have a frame defined at its center (as shown in the figure) and we know the poses of the face-centered frames for each of the faces (F_A is the frame of the top face on A, and F_B is the frame on the bottom face of B); the z-axis of the face frames points **outward** (as shown in the figure). We also have a frame G defined on the gripper, as shown on the left.



We know: ${}^{G}X^{B}$, ${}^{W}X^{A}$, ${}^{A}X^{F_A}$ and ${}^{B}X^{F_B}$

You can use $T_x(.) = \mathit{RigidTransform}(\mathit{MakeXRotation}(.))$ and similarly $T_y(.)$ or $T_z(.)$ in your answers.

(a) Give an expression for ${}^{W}X^{G_{place}}$, the rigid transform for the desired gripper placement, as shown on the right side of the figure.

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2 QP

2. In a problem set, we saw how to keep the end-effector position within a specified "box" in Cartesian coordinates. In this problem, we want to keep the end-effector position near a specified **plane** as well as within a specified box. Imagine that you want to write on a whiteboard or wipe a table. Assume the plane equation is available as a vector of 4 numbers $[n_x, n_y, n_z, d]$ ($\hat{n} = [n_x, n_y, n_z]$ is a unit normal) and the maximum distance allowed from the plane is given as $d_{\max} = 0.01$.

   Recall that the signed distance from the point $p = [p_x, p_y, p_z]$ to a plane is $\hat{n} \cdot p + d$; the distance is its absolute value. Assume p, the plane, and the translational rows of J_G are expressed in the same world frame, and that these rows give the velocity of the constrained end-effector point. And remember that the spatial velocity is $\begin{pmatrix} \omega \\ v \end{pmatrix}$.

   Here is the key function from the problem set.

```
def DiffIKQP_Wall(
    J_G: np.ndarray,
    V_G_desired: np.ndarray,
    q_now: np.ndarray,
    v_now: np.ndarray,
    p_now: np.ndarray,
) -> np.ndarray:
    prog = MathematicalProgram()
    v = prog.NewContinuousVariables(7, "joint_velocities")
    v_max = 3.0  # do not modify
    h = 4e-3  # do not modify
    lower_bound = np.array([-0.3, -1.0, 0.0])  # do not modify
    upper_bound = np.array([0.3, 1.0, 1.0])  # do not modify

    error = J_G.dot(v) - V_G_desired
    prog.AddCost(error.dot(error))

    prog.AddConstraint(le(v, v_max * np.ones(7)))
    prog.AddConstraint(ge(v, -v_max * np.ones(7)))
    prog.AddConstraint(le(p_now + h * J_G[3:6, :].dot(v), upper_bound))
    prog.AddConstraint(ge(p_now + h * J_G[3:6, :].dot(v), lower_bound))

    solver = SnoptSolver()
    result = solver.Solve(prog)
    if not (result.is_success()):
        raise ValueError("Could not find the optimal solution.")
    v_solution = result.GetSolution(v)
    return v_solution
```

(a) What **new** constraints would need to be added to the QP from the problem set? Express your answer as code to add. Assume a variable `plane` is available that holds the plane equation as an array and a variable `d_max` which is a scalar. You do not need to reproduce the whole function; just provide any new constraints.

(b) Provide a brief explanation of your answer.

3 Grasping

3. (a) We have seen in class that two point contacts with friction are insufficient to guarantee force closure on a three-dimensional body. As in class, we model point contact with no torsional friction. Assuming an antipodal grasp with two grasp points, a high friction coefficient and arbitrary squeezing forces, describe how such a grasp could fail to compensate for an applied force on the body, e.g. gravity.

(b) Now, consider the case in which the contacts are "patch" (also called "soft") contacts. Model such a contact as 3 point contacts with friction arranged in a triangle around a contact. This is a simple model of what we get when a (compliant) finger surface contacts an object surface and we have an area of contact. Explain how two antipodal surface areas can resist the applied forces in your counterexample above.

(c) Assume that we want to pick an object in one orientation and place it in a different orientation or place it close to other objects. When choosing antipodal grasps we heuristically check for collisions (with the point cloud) at the pickup point but ignore any collisions at the target pose. Assuming you know the pick pose ${}^{W}X^{O_{Pick}}$ and the place pose ${}^{W}X^{O_{Place}}$ how can you modify the point cloud so that the current grasp sampler produces grasps that are collision-free (with respect to the point cloud) for both the pick and place pose?

4 RRT

4. (a) Assuming a mobile robot wants to move from one side of a room to the other while avoiding collisions. In which setting do you expect RRT to struggle the most? Circle your answer.

   1. A cluttered room with many small obstacles scattered evenly around
   2. A cluttered room with obstacles scattered evenly around and a column in the center.
   3. An open room with a few large obstacles
   4. An open room with a wall in the middle such that there is a single narrow door to get from one side of the room to the other

   Briefly explain your answer.

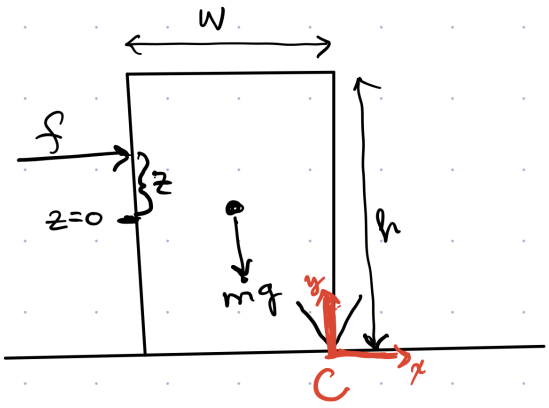
   (b) When searching for efficient paths, we have to beware of "shortest paths" since such paths move very close to obstacles, in fact, touching the obstacles. Consider two strategies for achieving some clearance:

   1. run RRT-Connect multiple times, return the path that minimizes (pathLength - alpha * minClearanceAlongPath) for some value of alpha.
   2. run RRT-Connect once but don't add to the tree any samples that are too close to obstacles (using some pre-defined threshold).

   Give some pros and cons for both of these strategies; be sure to include the impact on completeness.

5 Force control

5. Consider a block of height h and width w on a horizontal plane. Its mass is m and the coefficient of static friction is μ. We apply a purely horizontal force of magnitude f on the left side at a height of z relative to the center of mass, that is, at $z + h/2$ above the support surface. Note that $-h/2 < z < h/2$.



We want to understand what applied force will cause the block to slide or pivot around the bottom right corner C. Analyze the onset of motion as the force is increased quasistatically from rest. At incipient tipping, the support resultant acts at the bottom right corner. Use this limiting configuration for the torque calculation; subsequent sliding/tipping dynamics are outside the scope of this problem.

Recall that the (signed) magnitude of the cross-product $a \times b$ for planar vectors is $a_x b_y - a_y b_x$.

(a) Consider the torques **acting around the bottom right corner** (the potential pivot point).

i. What is the signed magnitude of the torque due to the applied force f?

ii. What is the signed magnitude of the torque due to the gravitational force?

iii. What is the signed magnitude torque due to the contact force (from the contact at the bottom right corner)?

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(b) What is the condition on f for tipping around the bottom right corner to occur? Hint: This should be a function of z. Write a mathematical expression and explain your reasoning.

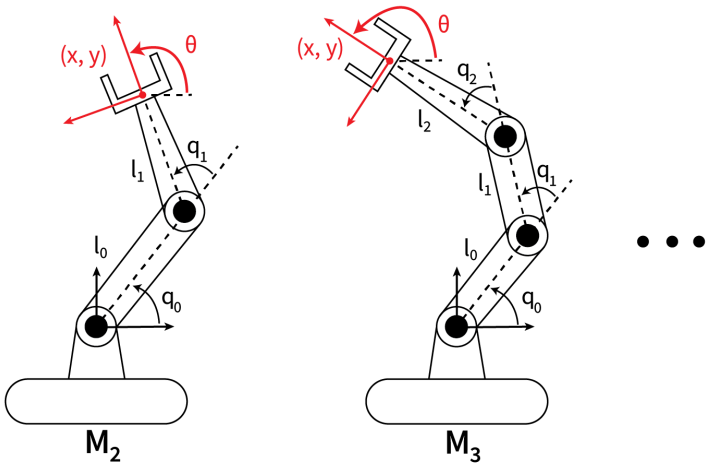
(c) What is the maximum (tangential) force due to friction?

(d) If we want sliding to begin before tipping as the force is increased from rest, how should we pick z (the height of the applied force)?

6 Jacobian

6. Consider a family of n link planar robots M_n (see the figure):

   - M_2: a 2-link manipulator with joint angles q_0, q_1, end-effector position x, y and end-effector orientation Θ.
   - M_3: a 3-link manipulator with joint angles q_0, q_1, q_2, end-effector position x, y and end-effector orientation Θ.
   - etc.



The forward kinematics of this class of robots can be written as:

$$\theta = q_{0:n-1}, \quad x = \sum_{i=0}^{n-1} l_i \cos(q_{0:i}), \quad y = \sum_{i=0}^{n-1} l_i \sin(q_{0:i}),$$

where $q_{0:k} = \sum_{i=0}^{k} q_i$. Let G be the end-effector (gripper) frame, whose center is at x, y in the world frame and whose orientation is θ in the world frame.

(a) Now, consider two alternative representations for the orientation of the gripper frame with respect to the world frame:

1. θ - a scalar angle
2. $(\cos(\theta), \sin(\theta))$ - a tuple of scalars between -1 and 1

We will refer to the **analytic** Jacobians for the first and second representation as $J^G_{a1,n}$ and $J^G_{a2,n}$, respectively, and the **geometric** Jacobians as $J^G_{g1,n}$ and $J^G_{g2,n}$, where n refers to links as in M_n.

Use the reduced planar geometric velocity $V^G = (\dot{\theta}, \dot{x}, \dot{y})^T$, expressed in world coordinates at G; G identifies the moving frame, not the expression frame. A full 3D spatial Jacobian would instead have six rows. Use analytic coordinates $(\theta, x, y)^T$ and $(\cos\theta, \sin\theta, x, y)^T$, respectively. The orientation-first ordering follows the Drake convention.

Hint: The matrix E requested below will have this structure: $\begin{bmatrix} M & 0 \\ 0 & I \end{bmatrix}$, for some unknown matrix M (which will depend on θ), so you don't need to compute $\partial x/\partial q$ and $\partial y/\partial q$.
Recall that the derivative of $\sin(x)$ is $\cos(x)$ and the derivative of $\cos(x)$ is $-\sin(x)$.

i. (2 points) What is the shape of the **geometric** $J^G_{g1,n}$ matrix for the scalar representation of rotation?

ii. (2 points) What is the shape of the **analytic** $J^G_{a1,n}$ matrix for the scalar representation of rotation?

iii. (3 points) Write the matrix E such that $J^G_{a1,2} = EJ^G_{g1,2}$.

iv. (2 points) What is the shape of the **geometric** $J^G_{g2,n}$ matrix for the tuple representation of rotation?

v. (2 points) What is the shape of the **analytic** $J^G_{a2,n}$ matrix for the tuple representation of rotation?

vi. (3 points) Write the matrix E such that $J^G_{a2,2} = EJ^G_{g2,2}$.

(b) Assume a non-singular configuration of the manipulator and a commanded spatial velocity $V^G = [\dot{\theta}, \dot{x}, \dot{y}]$. Let:

$$v_n = [J_n^G]^+ V^G$$

and

$$\hat{V_n^G} = J_n^G v_n$$

where n indicates the number of links and $[\cdot]^+$ denotes the pseudo-inverse. We'll assume geometric Jacobians for a scalar rotation representation, i.e. $J_n^G = J_{g1,n}^G$ from the previous part.

Assume $\text{rank}(J_n^G) = \min(3, n)$. For $n = 2$, take a command V^G outside the column space of J_2^G; this makes the intended unachievable-command case explicit. For $n = 3, 4$, the command is arbitrary. Select all true statements under these assumptions:

1. $\hat{V_n^G} = V^G$
2. $\hat{V_n^G} \neq V^G$, and v_n is the unique minimizer of $\min_v \|J_n^G v - V^G\|^2$.
3. v_n is the unique value for which $J_n^G v = V^G$
4. There are infinitely many v satisfying $J_n^G v = V^G$; $v_n = (J_n^G)^+ V^G$ is the unique minimum-norm member of this set.

i. (2 points) For $n = 2$ which of these statements hold? Explain briefly.

ii. (2 points) For $n = 3$ which of these statements hold? Explain briefly.

iii. (2 points) For $n = 4$ which of these statements hold? Explain briefly.

End of questions.